

Home Task:



Investigation 1 – The sieve of Eratosthenes

In this investigation you will determine what Eratosthenes' algorithm does.



	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

You will need 4 colors (preferably highlighters) to do this investigation. Start with a grid of numbers from 2 to 100, like the one shown here.

- 1 The first number in the list is 2, but do not highlight the 2. Highlight every 2nd number in the grid *after* 2 (all the multiples of 2).
- 2 At this point, the lowest number not yet highlighted is 3. Do not highlight the 3, but highlight every 3rd number in the grid *after* 3 (all the multiples of 3).
- 3 The next lowest number not yet highlighted is 5, so highlight all the multiples of 5.
- 4 Now the next lowest number not yet highlighted is 7, so highlight all the multiples of 7.
- 5 Take a look at all of the numbers that are not highlighted. Describe the pattern of these numbers. What kind of numbers are they?

